10 Unique Functionalities: How They Work in Our App

Smart Community Health Monitoring and Early Warning System for Water-Borne Diseases, Rural Northeast India

For each feature: who uses it, what they do, and what they see.

1. Camera-based Water Test Reader

Who uses it: ASHA worker or village volunteer.

- They dip a cheap test strip or H2S vial in the village spring water.
- They open the app, tap **Test Water**, and take a photo of the strip with a small colour card next to it.
- The app reads the colour and shows **Safe / Risky / Contaminated** in seconds.
- The result is saved with time, GPS and the water source name.

Why it helps: *No expensive sensors are needed. A phone camera does the job.*

2. Flood-triggered Pre-symptom Warning

Who uses it: District health officer and ASHA workers.

- The app automatically checks rainfall and flood data every day.
- If heavy rain or flooding is detected near a village, the app raises its risk level **before anyone falls sick**.
- ASHA workers get a message: *Heavy flood in your area. Check water sources and keep ORS ready.*

Why it helps: *Most systems only react after people are sick. This one warns early.*

3. Water-source Level Risk Map

Who uses it: Everyone, mainly officers and ASHA workers.

- Every spring, well, hand pump and tap is marked as a point on a map.
- Each point has a colour (**green, amber or red**) based on its latest test result, nearby rain or flood, and reports of sickness near it.
- Tap a red point to see its history: last test, last cleaning and nearby cases.

Why it helps: *Instead of saying 'Village X is risky', the app says exactly which source is the problem.*

4. Alert-to-Action Engine

Who uses it: Health officers and ASHA teams.

- When a village turns red, the app does not just show a colour.
- It creates a **task list**, for example: *Send 200 ORS packets and 50 chlorine tablets. Assign ASHA team 2. Visit the spring first.*
- Workers tick the tasks off as they finish. Officers see what is done and what is pending.

Why it helps: *An alert is useless unless someone acts. This turns the alert into clear steps.*

5. Sample Chain-of-Custody (QR + Hash)

Who uses it: ASHA worker, transport person, lab staff.

- When a water or stool sample is collected, the app creates a **QR code** and sticks it on the sample.
- Each handover (village, block, district lab) is scanned and logged with time and location.
- If the data is changed later, the app shows a mismatch.

Why it helps: *No more lost or delayed samples, and nobody can quietly edit records.*

6. Peer-to-peer Offline Sync

Who uses it: ASHA workers in hilly areas with no network.

- Reports and tests are saved on the phone when there is no internet.
- When two ASHA workers meet, their phones can share the latest data through Bluetooth or Wi-Fi Direct.
- Anyone who reaches network later uploads everything for the group.

Why it helps: *Data reaches the server even if no single phone has signal for days. For the hackathon, show a simpler version: save offline and sync later.*

7. Voice Symptom Triage (Local Languages)

Who uses it: Villagers and ASHA workers who find typing hard.

- The user taps the mic and says in their own language, *My child has loose motion and is very weak.*
- The app asks a few simple voice questions: *How many days? Is the child drinking water? Any sunken eyes?*
- It then gives guidance: **Give ORS now** or **Go to the health centre immediately**, and records the report.

Why it helps: *Important: this is guidance, not a diagnosis. The app always tells the user to see a health worker or doctor in serious cases.*

8. Early Signals from Non-medical Sources

Who uses it: The system itself, mostly in the background. Officers see the result.

- The app also tracks signs that appear before hospital cases: **ORS/zinc sales at shops, school absences, ORS stock finishing with ASHA workers**.
- If these jump suddenly in an area, the app shows a **Possible early outbreak signal** on the dashboard.

Why it helps: *These signs show up days before official case counts do.*

9. Outbreak What-if Simulator

Who uses it: District officers making decisions.

- The officer opens a village on the dashboard and tries options like: *What if we chlorinate this spring today?* or *What if we do nothing for 5 days?*
- The simulator shows an estimated difference in cases, for example as a simple graph with two lines.

Why it helps: *It makes the decision easier to explain and prioritise. The numbers are estimates, not promises, and should say so on screen.*

10. Cluster Detection for Sparse Data

Who uses it: The system, quietly. Officers see the result.

- The app watches reports coming from villages. If several cases appear close together in space and time, it marks them as a **cluster**.
- It works even with few reports, which is common in rural areas, and it can show **why** it flagged the area (for example, 9 cases in 3 days, normal is 2).

Why it helps: *It catches small local outbreaks early without needing big training data.*

How they connect (one-line story)

Flood warning (2)	**Map turns amber (3)**	**ASHA tests water with camera (1)**

Cluster confirmed (10)	Action tasks created (4)	Sample tracked to lab by QR (5)

Read left to right, then second row. Features 6 to 9 make the system stronger and are fine as future scope in the presentation.